

losing all three of its valence electrons ($4s^2$ $3d^1$) to form a +3 oxidation state. It lacks the ability to exhibit multiple states, unlike other transition metals, due to this highly stable +3 state.

56. The intense purple colour of potassium permanganate ($KMnO_4$) is due to Ligand-to-Metal Charge Transfer (LMCT). Specifically, it results from the excitation of an electron from an oxygen ($O^{2-}$) ligand orbital to a vacant d-orbital of manganese ($Mn^{7+}$), absorbing yellow-green light and showing purple.

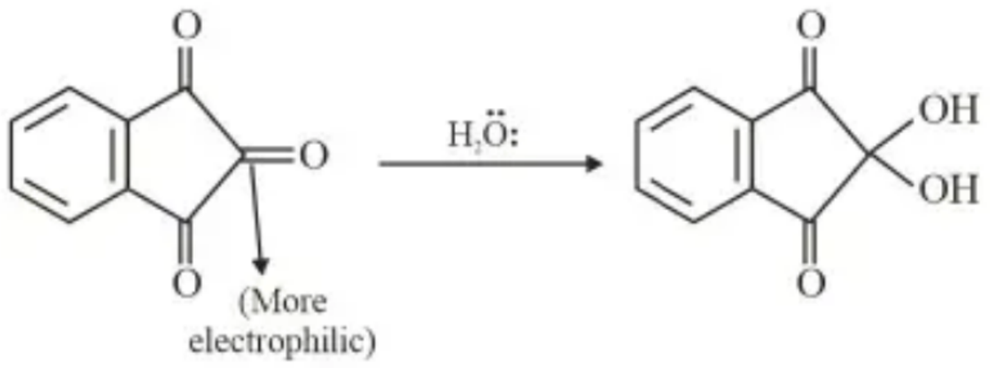
57. Based on the principles of inorganic chemistry and back-bonding, the lowest C–O bond order among common metal carbonyls is found in the $[V(CO)^6]^-$ anion.

58. Cisplatin is used for the treatment of cancer.
Yes, Vitamin B12 (cobalamin) contains cobalt (Co) as its central metal ion, not iron (Fe).
"Sodium EDTA (or Disodium EDTA) is used as a chelating agent to remove calcium and magnesium ions from hard water by forming water-soluble complexes, thus softening the water and inactivating the hardness-causing minerals."

59. $As^{+3}|Sb^{+3}|sn^{+2}$ are soluble in YAS|Yellow ammonium sulphide where as $Cd^{2+}|Cu^{2+}|Hg^{2+}|Pb^{2+}$ are insoluble in YAS Cds does not soluble due not form a soluble thiocomplex.

60. Iodoform test given by methyl ketone and methyl aldehydes only.

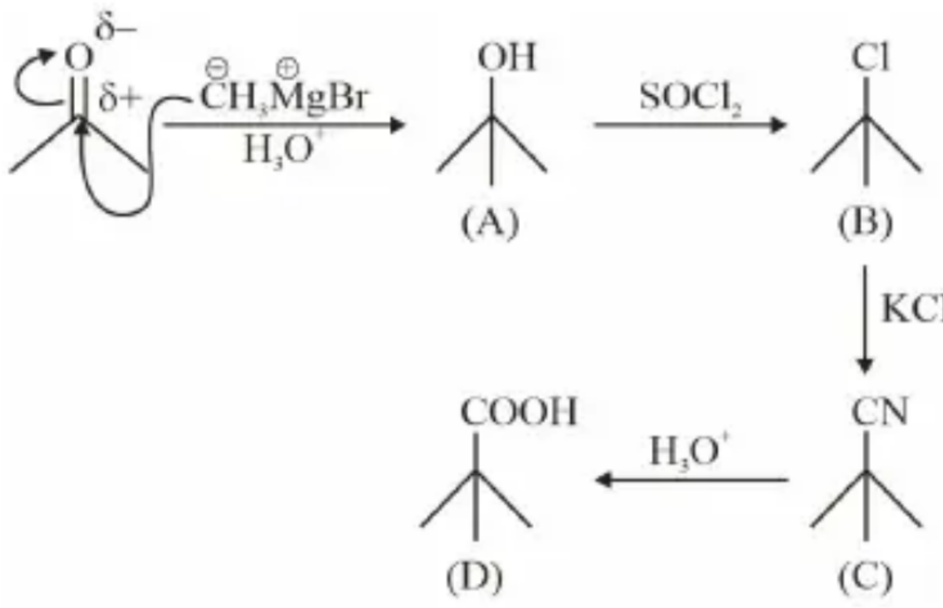
61.



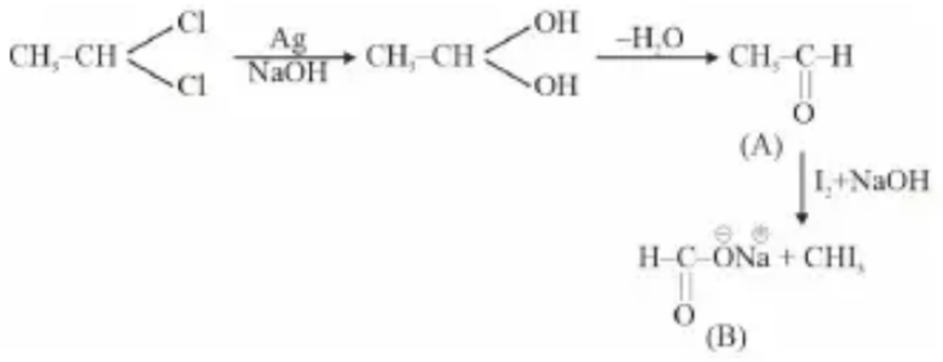

62. Friedel craft Acylation reaction

63. In Benzoate ion, equivalent resonance is present, but in phenoxide ion, non-equivalent resonance is present.

64.

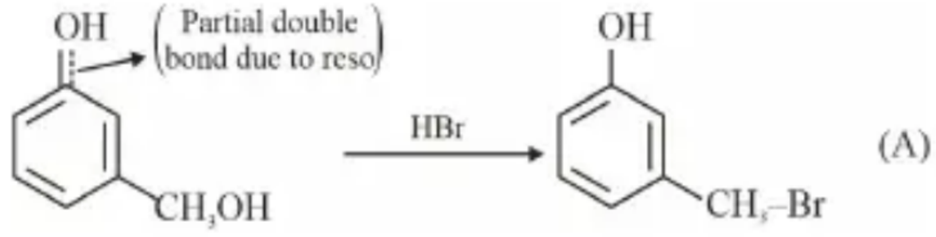



65.

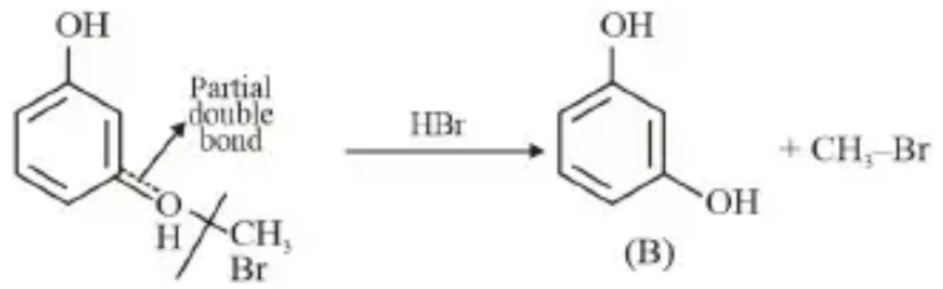



66.

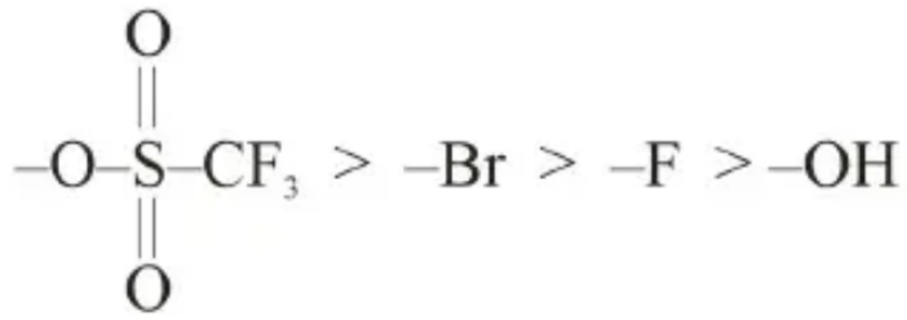
(i)




(ii)

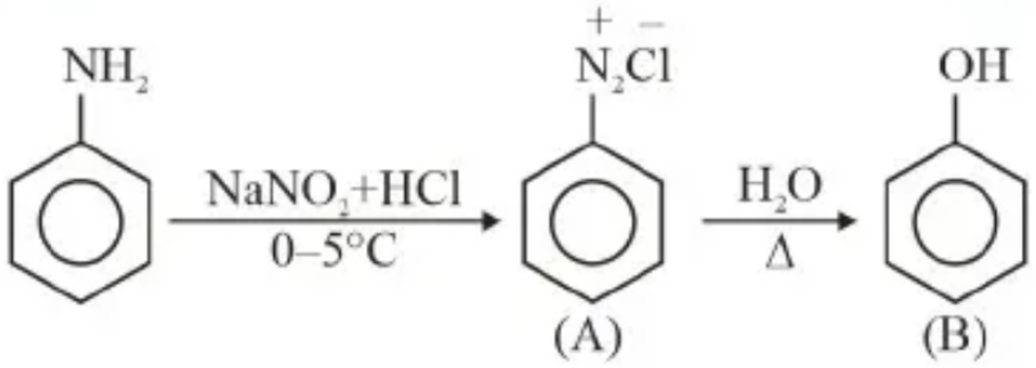



67. L.U. tendency :

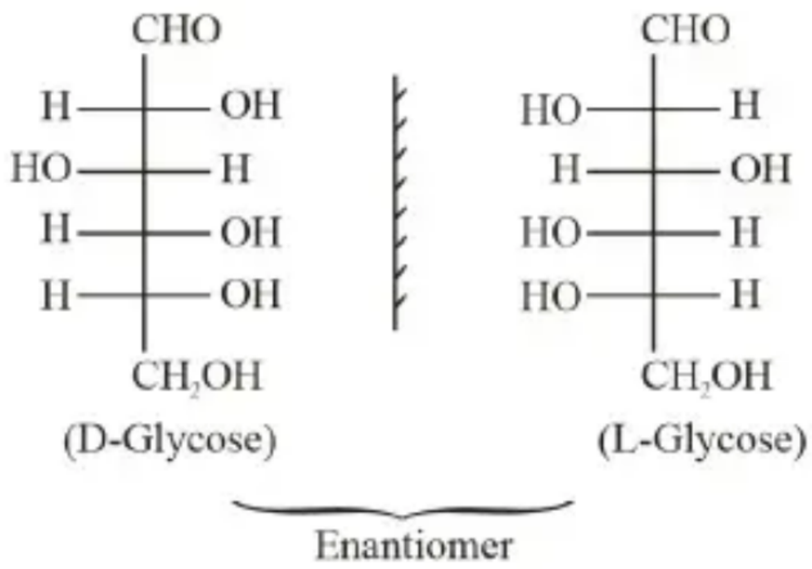



68. I, III, IV □ Aromatic
II □ Non-Aromatic
Basic Strength Order : IV > II > III > I
In conjugate acid of IV, equivalent resonance is present.

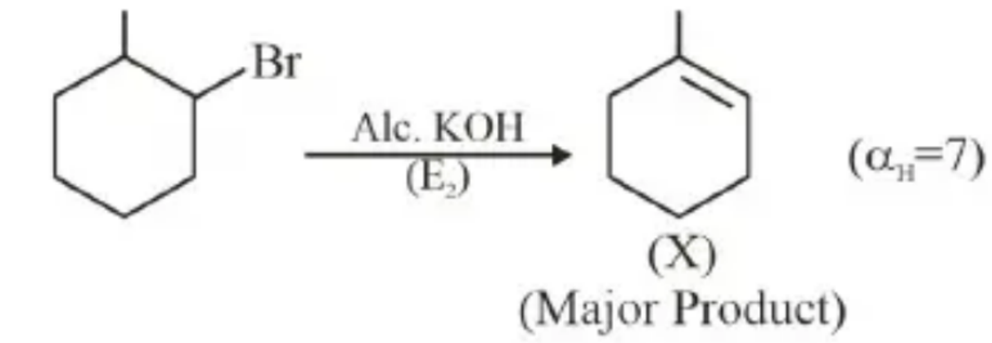
69.



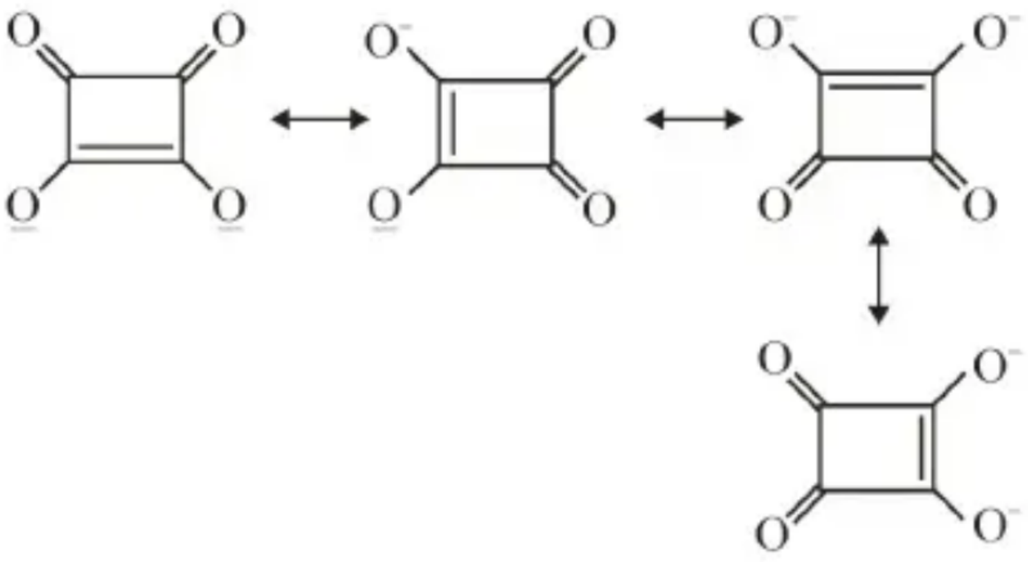

70.



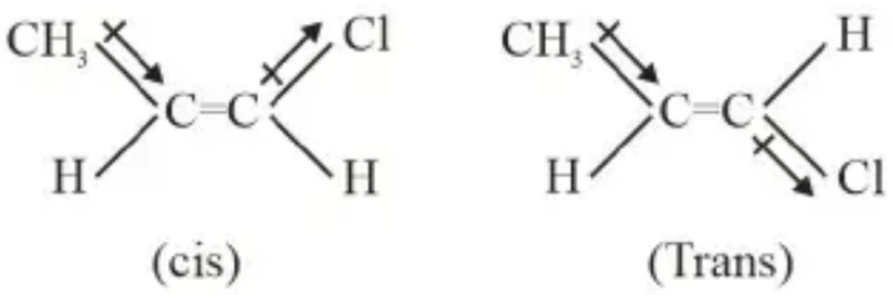

71.




72.




73.




$\mu$ : <

B.P. <

82. Hint :

$\Delta H_{ion} = \left|-13.6-(-11.3)\right|$

$2.3\frac{kcal}{mol.}$

85. $C_6H_{6(l)} + \frac{15}{2}O_{2(g)} \rightarrow 6CO_{2(g)} + 3H_2O_{(l)}$

0.1mol

U $\frac{327}{0.1}$ $3270\frac{kJ}{mole}$

$\Delta ng = -1.5$

$\Delta H = \Delta U + \Delta ngRT$

$(-3270) \frac{(-1.5)(8.314)300}{1000}$

$3273.74\frac{kJ}{mol.}$

86. $pOH = pKb + \log \frac{[Salt]}{[Base]}$

$= 4.8 + \log\frac{0.1 \quad 50}{0.01 \quad 50}$

$pOH = 5.8$

$pH = 14 - 5.8$

$= 8.2$

87. $kp \quad \frac{(p_{CO})^2}{(p_{CO_2})}$

$= \frac{8\times 8}{4}$

$= 16$

88. $CaCO_3 \rightarrow CaO + CO_2(g)$

Pure : $120 \times 10^3$ gm

$= \frac{120\times 10^3}{100}$

= 1200 mole

$n_{CO_2}$ = 1200 mole

Volume @ STP = $1200 \times 22.4$

$= 2.688 \times 10^4$ Lit

89. Theory Based

90. $Ag_2C_2O_4 \rightleftharpoons 2Ag \quad C_2O_4^{2}$

– 2S S

$2S = 2.2 \times 10^{-4}$

$S = 1.1 \times 10^{-4}$

$k_{SP} = (2S)^2.S$

$= 4S^3$

$= 4 \times (1.1 \times 10^{-4})^3$

$= 5.3 \times 10^{-12}$